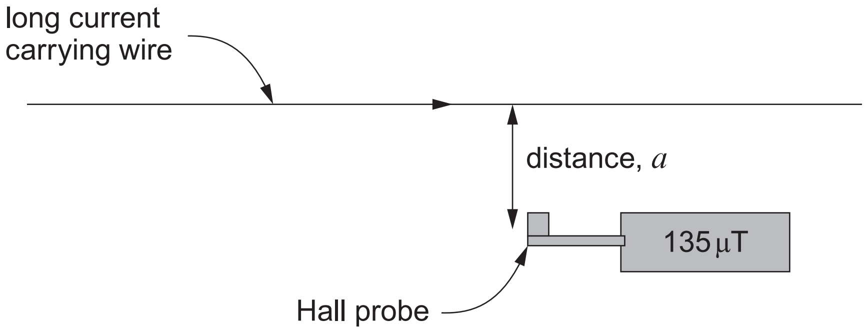
Examiner only

5. Michael carries out an experiment to determine how the magnetic field, B, due to a long current carrying wire varies with distance, a, from the wire.



His results are recorded in a table and a graph of $\ln(B)$ against $\ln(a)$ is plotted using the corrected B-field.

a/m	Measured B-field/T	Corrected B-field, B/T	$\ln(a/\text{m})$	$\ln(B/\text{T})$
0.010	0.000 529	0.000 499	−4.61	−7.60
0.020	0.000 279	0.000 249	−3.91	−8.30
0.040	0.000 157	....	....	....
0.080	0.000 092	0.000 062	−2.53	−9.69
0.160	0.000 063	0.000 033	−1.83	−10.32
0.300	0.000 046	0.000 016	−1.20	−11.04

(a) Explain why the corrected B-field, B must be obtained. [1]

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(b) **Complete the table** above and **plot the missing point** on the graph opposite (no error bars are required). [3]

(c) The gradient of the steepest line is −1.03. Calculate the gradient of the least steep line. [2]

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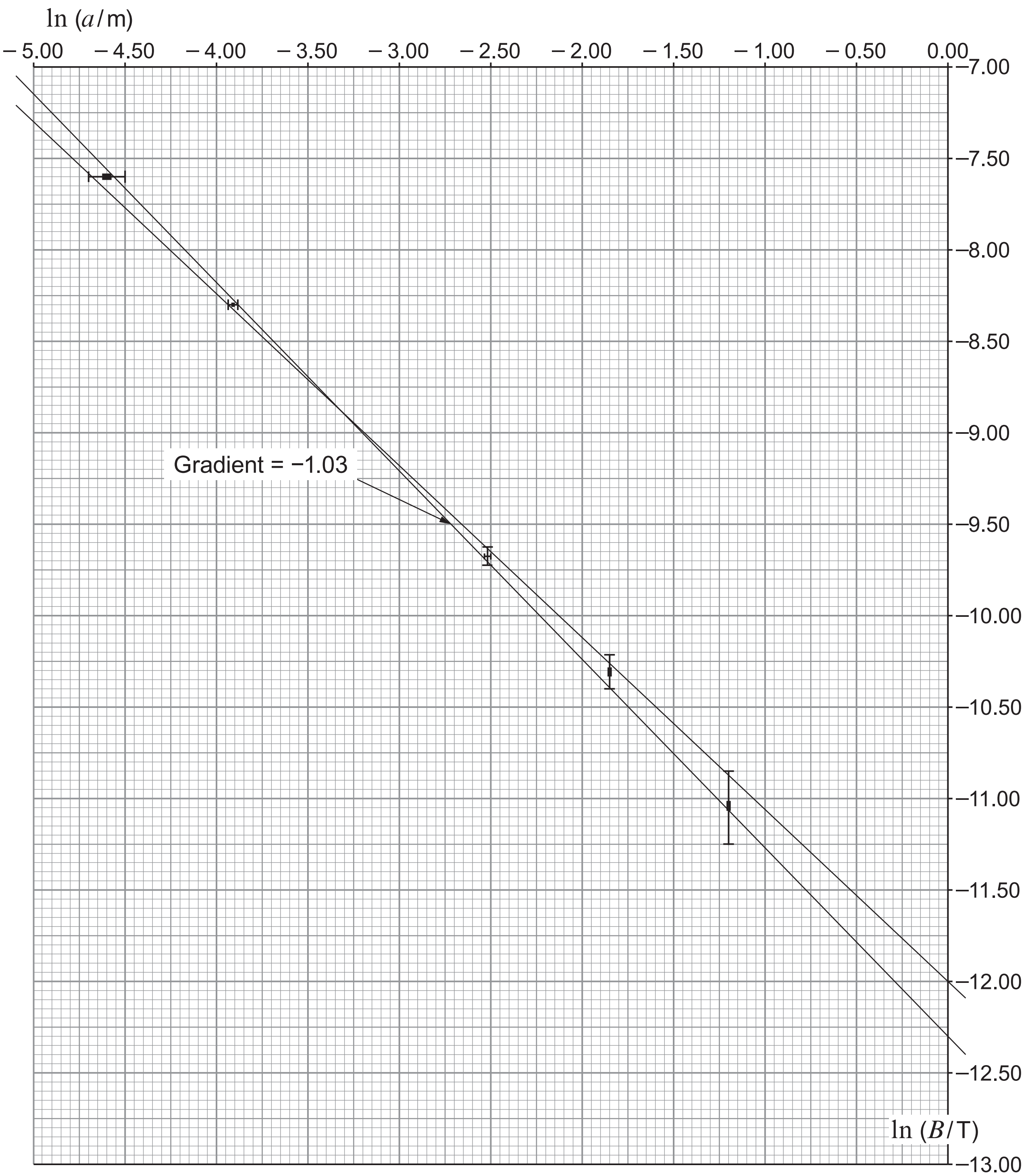

(d) Use the equation $B = \dfrac{\mu_0 I}{2\pi a}$ to show that the expected mean gradient is −1 and that the expected $\ln(B)$ intercept is $\ln\left(\dfrac{\mu_0 I}{2\pi}\right)$. [2]

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(e) Explain to what extent the graph confirms the relationship $B = \dfrac{\mu_0 I}{2\pi a}$. [3]

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(f) Calculate the current in the wire along with its **absolute** uncertainty. [5]

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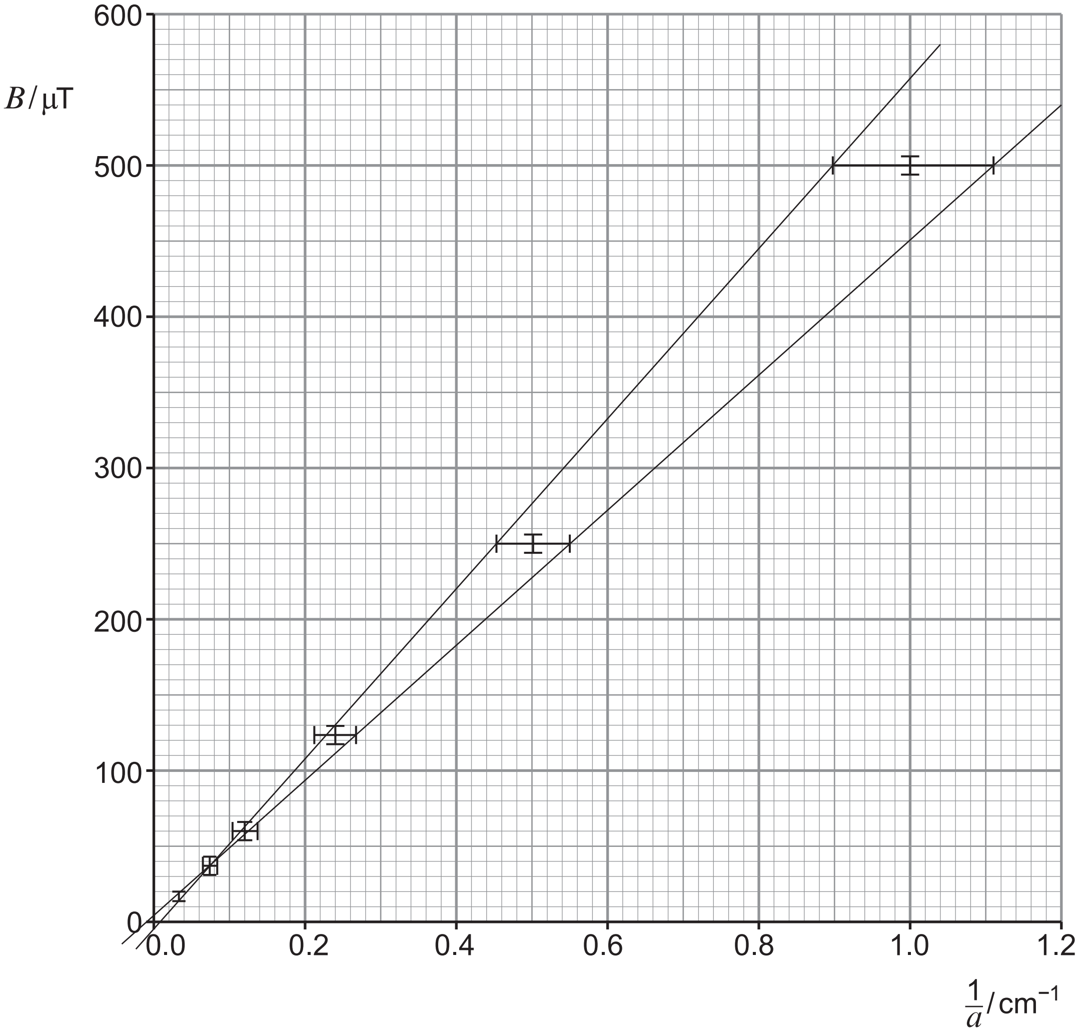
(g) Bethan analyses Michael's data by plotting a graph of B against $\frac{1}{a}$.



Explain briefly why Michael's log graph is more suitable than Bethan's graph for this data set (no calculations are required). [2]

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